



The Food Group Student Data Challenge

MUDAC 2026



Need And Demand Are Not The Same

1 min

- Need is the help that a household must have.
- Demand is the help that a household asks for.
- A visit measures demand. A visit does not measure need.
- The Food Group allocates resources by visits. Allocation therefore follows demand.
- An area that is easy to reach makes visits. It draws resources.
- **Your task: show where need and demand agree. Show where they differ. Explain the cause.**

27%

of working-age adults are food insecure

17%

of working-age adults use charitable food

1 in 3+

users of charitable food are food secure

Source: Urban Institute, Well-Being and Basic Needs Survey



The Data For This Challenge

1 min

1

Four datasets come from The Food Group

Food shelf activity. Call center logs. MarketBucks participation. Organization records.

2

One public dataset is mandatory

Feeding America Map the Meal Gap. You must request it yourself.

3

All other data is your choice

You select the source. You defend the choice.

4

Two documents support the data

The Glossary of Terms. The Data Dictionary.



Food Shelf Activity

2 min

- Each row is one food shelf location for one month. The data starts in 2022.
- The data reports three metrics: visits, pounds distributed, and individuals served.
- One visit is one trip by one household. A visit is not a unique person.
- The data splits individuals served into children, seniors, and adults.
- **The three metrics do not always move together. Report all three.**



Call Center Logs

1.5 min

- Each record is one call.
- Each record gives the city, the county, the reason for the call, and the referral destination.
- A destination is a named food shelf, SNAP enrollment, or another resource.
- SNAP is the Supplemental Nutrition Assistance Program.
- **You define a food-related call. You defend that definition.**



Marketbucks Participation

1.5 min

- MarketBucks helps SNAP and SUN Bucks customers to buy more food at farmers markets.
- The data groups participation records by location and by time period.
- Many markets open for the summer months only. New markets also appear.
- **Use only the geographies and the time periods where the data sources overlap.**



Organization Records

1.5 min

- This is the master record for all partner organizations.
- It covers food banks, food shelves, and agencies.
- It holds location information and status fields.
- **Filter by organization type first. The table holds more than food shelves.**
- Check the status fields before you aggregate a location.
- This file makes the distance questions and the geography questions possible.



Feeding America Map the Meal Gap

3 min

- Map the Meal Gap estimates food insecurity for each county.
- The 2025 edition uses 2023 data. The 2026 edition uses 2024 data.
- Question 1 needs this source. Question 6 also needs it.
- We cannot give you the data. Feeding America controls the distribution.
- **Request the data early. An approval delay can stop your work.**

You must cite this source

The funds of Feeding America depend on recorded use and citations.



Find Your Own Data

3 min

- Other public data is welcome. The search is part of the challenge.
- Look at prices, income, poverty, population, employment, SNAP enrollment, and food retail access.
- Grain is the level of detail of one record.
- No dataset fits perfectly. Different geographies, different grains, and publication delays are normal.
- You must make trade-offs. The judges will ask you to defend them.

Welcome to the real world.

Record four items for each source

- 1 The publisher and the exact series
- 2 The geographic grain: county, tract, or region
- 3 The time grain: monthly, annual, or other
- 4 The reason for this source



The Two Supporting Documents

1 min

● The Glossary of Terms

The glossary defines the terms of this challenge. Examples are need, demand, food bank, and food shelf. Use these definitions in your report. The judges read your work with the same definitions.

● The Data Dictionary

The data dictionary defines each field in each dataset from The Food Group. Check the field meaning and the grain before you join two tables.

Read both documents first. Most early errors are definition errors.



The Questions For Each Division

2 min

Question	Novice	Undergrad	Graduate
1. Need estimates against observed activity	Yes	Yes	Yes
2. Calls as a signal	Yes	-	-
3. Farmers markets and food shelves	Yes	-	-
4. Food prices and food shelf activity	-	Yes	Yes
5. SNAP enrollment and food shelf use	-	Yes	Yes
6. A divergence measure for need and demand	-	Yes	Yes
7. A forecast of August activity	-	Yes	Yes
8. Low-income, low-access areas and network reach	-	-	Yes
Total questions	3	5	6

Scope rules

Question 1 is the base for all divisions.

Question 6 and Question 7 have a graduate addendum. Graduate teams answer both. Undergraduate teams answer neither.

The judges score only the questions of your division. Every dataset is open to every division.



One Rule Can End Your Competition

30 sec

- Question 7 asks for a forecast of August for Minnesota counties.
- August is in the past. Your data stops at July.
- Do not look for August data. Do not use August data from any source.

August data disqualifies your team.



The Use Of AI

1.5 min



AI tools are permitted

Use them. They are part of this work now.



The judges ask for your reasons

Expect a question about each choice: the data, the method, and the model.



"The AI suggested it" is not a reason

Know the source of each number in your report.



Do not present a result that you cannot explain

An unexplained result costs you more than an absent one.



How To Present

1.5 min

- Each team member must prepare to speak.
- Prepare fully. Do not read from a script.
- Know your sources, your assumptions, and your limits.
- Expect a question about any slide.

The judges ask

Where did this number come from?

Why this source and not another?

What would change your answer?

What does The Food Group do differently on Monday?



What Wins

1 min

A method you can explain often beats a complex one.

A model that staff can use

The Food Group must follow your logic. If the staff cannot follow it, they cannot defend your recommendation to a funder.

Connect each result to a decision

How to allocate resources. Where a new food shelf is needed. How to argue for a policy. How to measure results.